Q1

Data; any information a subject × recorded information

subject; what's being experiment on ✓

variable; the name // term for the data collected e.g. height, age ✓

EDA; Exploratory Data Analysis → viewing a dataset and

① Wrangling ✓ ② Summarising ✓

③ Data ✓ ___?___

④ Data ✓ ___?___

IDA ✓

Data background

→ where

→ How

→ who

→ Design of Experiment (w/ control)

Data Structure ✓

→ How dataset is organised

→ Spreadsheet

→ media

→ csv?

Q2

A) discrete ✓ B) Nominal ✓ C) Discrete ✓ D) Ordinal ✓

E) Discrete × Qualitative nominal

• Because it is a label that identifies someone

↳ NOT ALL QUALITATIVE DATA IS CATEGORICAL

Q3

• Population ⇒ everyone ✓ • sample ⇒ subset of population ✓ being observed

Q4

tidy dataset → columns = variables
rows = observation / ~~values~~
cells = values

Graphical Summaries

Q5

a) Bar graph ✓
b) Boxplot or Histogram ✓
c) Scatterplot ✓
d) Boxplot ?? ✓ 1 Quantitative and 1 Qualitative ~~Bar graph?~~
e) Bar graph ✓

Q6 (relationship)

hours studied → discrete

exam mark → ordinal

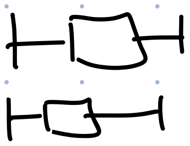
→ ∴ Scatter plot ✓

Q7

eye colour is nominal categorical (Qualitative) data ∴ the use class intervals width is useless → no x-axis

→ use bar graph ✓

Q8



1. Median
2. IQR
3. Q1 vs Q3
4. Shape of Data

Q9

• Distribution

• shape, center, skew, cluster

Q10

$Q1 = 12$
$Q_3 = 20$

$IQR = 8$

$LT = 12 - 1.5 \times 8$
$= 0$

$UT = 20 + 1.5 \times 8$
$= 32$

Q11

any value < LT ∪ UT <

Q12 ?

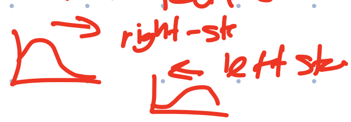
boxplot → shows based on Quantiles ~~to~~ measure of spread & tendency
histogram → shows distribution ⟹ more granular!

boxplot → compact summary, histogram → distribution shape

Q13

→ high income earner?

skew refers to where the tail of the data stretch



Q14.

association

• strong positive $\boxed{\text{correlation}}$ ✓

• only correlation if linear relationship

why not correlation (i.e it ∵ of relationship) observable data

Q15

• Max values are very high → outliers are moderately frequent.

R

Q16

• str( ) ? structure ( ) ⇒ str( ) → shows STRUCTURE/ TYPES

• head( ) → 1st 4 lines of data

• Tail( ) → last 4 lines of data

Q18

? Tidyverse = collection of R packages for tidy data analysis, wrangling, graphing.